

EFFECT OF NICOTINE ON THE MORPHOLOGY OF ADRENAL CORTEX

Ishrat S. Shokh

Department of Anatomy, Sindh Medical College, Karachi, Pakistan.

ABSTRACT

The morphological effects of nicotine on the adrenal cortex have been observed in this study to get supportive evidence whether nicotine has some effect on the adrenal cortex; as regards the morphology.

Animals were divided into three groups, which comprised group A control group, group B treated group and group C placebo group. Among these animals half of the animals from each group were sacrificed after 40 days and the remaining half after 60 days of treatment. Group A was control group, group B was given Nicotine Sigma 1 mg/kg body weight daily intra peritoneally for 60 days and B₂ for 60 days. Group C and C₁, were given the same quantity of saline intra peritoneally daily for same period.

On autopsy adrenals were removed and fixed in normal saline. 5 micron thick sections in paraffin and 10 micron thick frozen sections were studied for the cortical zones, cell size in each zone cell size in each zone and among of lipid in cortical cells. It was observed that in treated group there was no change in B₁ group but there was a marked hypertrophy of fasciculata cells with increased width of zona fasciculata.

To conclude, the present study supports the view that nicotine has stimulant effect on the adrenal cortex and prolonged administration of the drug can induce morphological changes in fasciculata cells.

INTRODUCTION

Smoking is one of the most ancient and wide spread menaces in the history of mankind. Nicotine is the active ingredient of that vast chemical complex known as "tobacco" and is responsible for major part of effects of smoking. It is a liquid alkaloid from the dried leaves of Tobacco plant. Epidemiological studies link nicotine with certain diseases, especially respiratory and cardiovascular diseases. Prolong use of nicotine induces a rise in blood pressure and coronary heart disease (Bein *et al.*, 1978). Nicotine stimulates prostaglandins synthesis (Wennwalm, 1978), increases plasma cortisol and growth hormone concentrations and influences endocrine system of body through two very sensitive endocrine mechanisms, that of pituitary and adrenal gland (Silvette, 1961). Administration of nicotine by intra peritoneal injections or through inhalation of tobacco smoke induces increased

plasma levels of adrenocortical glucocorticoids in different animals and in man (Kershbaum and Shafide, 1968).

Adrenal is an important endocrine gland of dual character.

Physiological action of adrenal hormones influences all body systems and is vital in many. Experimental removal or damage to adrenal cortex gives fatal results in animals and in humans. Because of physiological importance of adrenal cortical hormones and their broad use as a therapeutic agent, knowledge of effects of nicotine on the gland is of considerable interest and importance in the light of its wide spread use in the form of smoking.

There is some evidence which suggests that increased corticosteroid secretion is due to enhanced corticotrophin release resulting from a nicotine induced increase in sympathetic and catecholamine activity. Vogt (1951) reported that nicotine had no direct action on cortical tissue and that any increase in adrenocortical secretion in intact animals would have to be attributed to an increase in epinephrine level in circulation.

Cam *et al.*, (1979) has demonstrated that *in vivo* injection of nicotine induces increased plasma levels of corticosterone in normal rats but not in hypophysectomized animals indicating that adrenocortical response was mediated through ACTH release.

In contrast to previous evidence of Vogt (1951); it has been observed by Rubin and Warner (1975) that nicotine is capable of directly stimulating isolated adrenocorticoid cells of the cat. The stimulant effect is observable in micro molar concentrations and is like the physiological stimulus. Not only nicotine is able to stimulate cortical cells directly but it enhances the steroidogenic effect of ACTH. The interaction between nicotine and ACTH appears to be additive rather than synergistic. This conclusion is based upon the fact that the steroid response to ACTH plus nicotine approached the sum of their separate responses.

All the above mentioned observations are based on bio-chemical changes. No study has been carried out to observe the effects of nicotine on the morphology of adrenal cortex *in vivo*.

The present study was designed to determine the effect of nicotine on the morphology of adrenal cortex.

MATERIALS AND METHODS

Thirty six adult, male albino rats weighing 200-300 grams were selected and arranged in three equal groups.

Group A: This group consisted of 12 animals and served as control to treated and placebo groups. This group remained untreated.

Group B: This group also had 12 animals. The animals were given nicotine (sigma) prepared in normal saline interperitonally for 60 days.

Group C: This group likewise consisted of 12 animals and they were given the same quantity of normal saline interperitonally for 60 days.

DRUG AND DOSAGE

Nicotine sigma contains 1 grm. of drug/one millilitre. Dose of the drug was calculated 1 mg/kg body weight in the light of previous experiments (Seifert *et al.*, 1984).

Six animals from each group were sacrificed after 40 days and the remaining after 60 days of treatment. Guillotine method was used to avoid stress of anaesthesia. Both the adrenal glands were removed and fixed in normal saline.

TISSUE TREATMENT

1. One half of the gland was placed in 70% alcohol over night and then treated by ascending order of concentration of alcohol. After infiltration blocks were prepared. Tissue was placed in each paraffine block in such a position the horizontal sections from the middle of the gland could be obtained.

Five micron thick sections were cut on rotary microtome. Four sections from each block were taken on acid cleaned slides. Slides were dried on hot plate at 37° C leaving them over night on the plate.

2. Other half of each gland was cut as 10 micron thick frozen sections on freezing microtom. The sections were received in distilled water, then transferred to clean slides and stained.

STAINING

1. Paraffine sections were stained by haematoxylin and eosin stain.
2. Frozen sections were stained by sudan Block-B.

MICROSCOPIC EXAMINATION

Paraffin sections were studied for (1) measurement of cortical zones and for (2) measurement of cells from each zone by ocular micrometer.

MEASUREMENT OF ZONES

The measurement was done under 10 x objective lens. For this measurement, each section was divided into four quadrants by imaginary lines and two measurements from each quadrant were done to get the accurate measurement. The mean of all these were calculated in each slide.

SIZE OF THE CELLS

Twenty five cells from each zone of adrenal cortex were measured under 40 x objective lens. An effort was made to obtain representative sample

by measuring cells from different fields in all the selected sections of one gland.

METHODS OF CALIBRATION OF OCULAR MICROMETER

The stage micrometer scale with 2 mm length and equally divided in 200 parts with each division of 10 microns. The ocular micrometer was coincided with the stage micrometer and it was found that one division of ocular micrometer is equal to 580 m. Hence one division of ocular micrometer is equal to 11.6 m under 10 x objective lens of the particular microscope used.

Under 40 x objective, one division of ocular micrometer equals 3 m.

OBSERVATION AND RESULTS

The following observations were made and recorded.

1. The width of zones of adrenal cortex.
2. The diameter of the cells in microns in each zone of adrenal cortex.
3. The lipid contents in the cortical cells.

GROUP A (CONTROL GROUP)

Sub-Group A₁

The group was sacrificed after 40 days. The mean diameter of cells in zona glomerulosa of the adrenal cortex was 9.28 ± 0.37 . In zona fasciculata it was 14.63 ± 1.8 microns and in zone reticularis it was 8.01 ± 0.27 microns (table 2).

Sub-Group A₂

This group was sacrificed after 60 days. The means width of zona glomerulosa was 62.13 ± 1.36 microns, of zona fasciculata it was 321.54 ± 15.52 microns and that of zona reticularis it was 195.161 ± 11.06 microns.

The mean diameter of cortical (Table 1) cells in zona glomerulosa was 9.37 ± 0.14 microns. in zona fasciculata it was 14.76 ± 15.52 microns and in zona reticularis it was 8.37 ± 0.43 microns (Table 2).

Group B (Treated Group)

Sub-Group B₁

This group was treated for 40 days. The mean width of zona glomerulosa

Table 1 Width of Cortical Zones in Microns

Cortical Zones	A-1	B-1	* P value	C-1	*P value
Zona glomerulosa	62.85 ±2.25	62.23 ±1.68	>0.05	58.43 ±0.27	>0.05
Zona fasciculata	335.26 ±8.07	373.07 ±17.92	>0.05	351.45 ±14.62	>0.005
Zona reticularis	170.77 ±10.02	177.04 ±9.16	>0.05	181.03 ±3.13	>0.05

* = as compared to A1 group.

Table 2 Diameter of the Cells in Microns

Cortical Zones	A-1	B-1	* P value	C-1	*P value
Zona glomerulosa	9.28 ±0.31	9.2 ±0.31	>0.05	7.88 ±0.13	>0.05
Zona fasciculata	14.63 ±0.51	6.28 ±0.38	>0.05	14.55 ±0.1	>0.05
Zona reticularis	8.012 ±0.27	8.78 ±0.22	>0.05	7.56 ±0.17	>0.05

* = as compared to A1.

was 62.23 ± 1.68 microns. It was decreased as compared to control group but the difference was not statistically significant.

The mean width of zona fasciculata was 373.07 ± 17.92 microns. It was increased as compared to control group and the difference was statistically significant (Table 1).

The mean width of zona reticularis was 177.04 ± 9.16 microns. It was increased as compared to control group but statistically was insignificant.

The mean diameter of cells in zona glomerulosa was 9.2 ± 0.31 microns. Minor difference as compared to control was insignificant. The mean diameter of the cell in zona fasciculata was 16.28 ± 0.38 microns. It was increased as compared to control group but the difference was statistically insignificant (Table 2).

Sub-Group B₂

This group was treated for 60 days.

The mean width of zona glomerulosa was 59.06 ± 0.65 microns. It was less as compared to control and the difference was insignificant. Mean width of zona fasciculata was 377.13 ± 10.0 microns. It was increased as compared to control and the difference was statistically significant (Table 3).

Mean width of zona reticularis was 187.11 ± 0.08 microns. It was less as compared to control and difference was insignificant.

The cell diameter in zona glomerulosa was 8.35 ± 0.26 microns, less than control and the difference was statistically insignificant (Table 2).

The mean cell size in zona fasciculata was 16.06 ± 0.42 microns. It was increased as compared to control and was statistically significant (Table 6). The cell size reticularis was 8.07 ± 0.02 microns. It was more as compared to control group but the difference was insignificant.

Group C — Placebo group

Sub-Group C₁

The mean width of zona glomerulosa decreased as compared to control group. The difference was insignificant (Table 3). The mean width of zona fasciculata was 351.45 ± 15.6 microns. It was increased as compared to control group and the difference was insignificant. The mean width of zona reticularis was 181.03 ± 3.13 microns. It was increased as compared to control group and the difference was insignificant.

The mean diameter of cells of zona glomerulosa was 7.88 ± 0.13 microns. It was less than control group and difference was insignificant.

The mean diameter of cells in zona fasciculata was 14.55 ± 0.01 microns. It was less than control but no significant. The mean diameter of cells in the zona reticularis 7.56 ± 0.17 microns. The difference is in significant.

Sub-Group C₂

The mean width of zona glomerulosa was 61.09 ± 1.44 microns which was less than control group and difference was insignificant.

The mean diameter of zona fasciculata was 345.67 ± 14.44 microns. It was increased as compared to control but the difference was statistically insignificant.

The mean width of zona reticularis was 188.58 ± 0.30 microns. It was less as compared to control group but the difference was statistically insignificantly.

The main diameter of cells in zona glomerulosa was 9.23 ± 0.34 microns. It was less than control group but the difference was statistically insignificant.

The mean diameter of cells in zona fasciculata was 15.03 ± 0.30 microns. It was more as compared to control and difference was statistically insignificant.

The mean diameter of cells in zone reticularis was 7.79 ± 0.24 microns. It was less the control and difference was statistically insignificant (Table 4).

LIPID CONTENTS (SECOND HALF OF THE GLAND)

The cells of the zona glomerulosa were rich in lipids and distribution of lipids is more or less constant. A few lipid free cells could be demonstrated.

In outer fasciculata the lipid droplets were extremely but definitely less than glomerulosa. The size of lipid glomerulosa was variable.

In zona reticularis, the cells are in marked contrast with those of other zones. They are small arranged in alveoli with large nuclei. Their cytoplasm is poor in stainable lipids.

Group B

Intracellular lipid globules remained almost constant in zona glomerulosa, and showed no apparent change in treated group when compared with the control group.

Zona fasciculata showed decrease in size and distribution of lipid globules in the cytoplasm of the cells. The cells showed lipid depletion with little increase in their size. Zona reticularis showed no change in intracellular lipid.

Group C

The cells showed no change in staining characteristics. Intercellular lipid droplets were the same in size and distribution as compared to control group.

Table 3 Width of Cortical Zones in Microns

Cortical Zones	A-2	B-2	* P value	C-12	*P value
Zona glomerulosa	62.13 ±1.36	59.06 ±0.65	>0.05	61.09 ±1.44	>0.05
Zona fasciculata	321.54 ±15.52	377.13 ±10.00	>0.05	345.67 ±14.44	>0.05
Zona reticularis	195-16. ±11.06	187.11 ±5.23	>0.05	188.55 ±7.46	>0.05

* = as compared to A2.

Table 4 Cell Size in Cortical Zone in Microns

Cortical Zones	A-2	B-2	* P value	C-2	*P value
Zona glomerulosa	9.37 ±0.14	8.35 ±0.26	>0.05	9.22 ±0.34	>0.05
Zona fasciculata	14.76 ±0.18	16.06 ±0.42	>0.05	15.02 ±0.3	>0.05
Zona reticularis	8.37 ±0.43	8.07 ±0.08	>0.05	7.79 ±0.24	>0.05

* = as compared to A2 group.

DISCUSSION

In this study there was no significant change in the cellular arrangement of cells size of cortical zones. Width of all the zones remained unaltered in short term treated group. In long term treated group of animals the zona glomerulosa showed no change either in the width of whole zone or in cell diameter. This confirmed the previous finding of Karsham and Shafide (1968) which suggests that zone glomerulosa and its products (aldosterone) is neither much influenced by ACTH nor by nicotine and the only change is in glucocorticoid secretions taking place which is related to fasciculata and reticularis. Zona fasciculata showed significant changes in cell diameter and in the width of the whole zone. There was a marked increase in the width of zona fasciculata and also an increase in overall dimensions of individual cells elements of parenchyma.

It also appeared that there was some stimulus acting on the fasciculata cells and in chronic stimulation both zona fasciculata and part of reticularis hypertrophy (Copen Laver *et al.*, 1978). Here hyperthrophy of fasciculata has occurred which proves that nicotine stimulates glucocorticoid production from adrenal cortex.

In the cells of zona fasciculata there is an extensive decrease in size of lipid droplets and distribution of lipids.

It has been established that the cellulose contents of lipid with a relative change in the size of cells reflect the state of activity of adrenal cells. Lipid depletion accompanied by hypertrophy of the cells indicate hyperactivity state of cells.

REFERENCES

- Ashmore, J. and Morgan, D., (1967). *Metabolic Effects of Adrenal Glucocorticoids Hormones, Carbohydrate, Protein, Lipid and Nucleic and Metabolism, in Adrenal Cortex*, edited by A.B. Eisenstein, Boston, Little Brown.
- Bird, C.E. and Clark, A.F., (1973). *The Adrenals in the Systemic Endocrinology*, edited by G. Ezrin by G. Ezrin, New York, Herper and Row.
- Bowman, E.R., (1962). Studies on the excretion of 5-(3-Pyridyl)-tetrahydrofuranon — 2 and its inter mediary role in the metabolism of nicotine, *J. Pharmacol. Exp. Ther.*, 135:306-11.
- Cam, G.R., Bassett, J.R. and Cairncross, K.O., (1979). The action of nicotine on the pituitary adrenal cortical axis, *Arch. Int. Pharmacodyn.*, 237 : 49-66.
- Coute-Devolx, B. Oliver, C., Giruad, P. and Gillioz, P., (1980). Effects of nicotine on *in vivo* secretion of malano-corticotrophic hormone in the rat, *Life Sci.*, 28 : 1067-73.

- Crossland, J., (1980). Lewis's *Pharmacology*. 5th ed., Churchill Livingstone Edinburgh.
- Deane, H.W., (1958). Intracellular lipids; their detection and significance in: *Frontier in Cytology*, edited by S.L. Palay. New Haven, Yale University Press.
- Dobbie, J.W. and Symington, T., (1966). The Human adrenal gland with special reference to vasculature, *J. Endocrinol.*, 34 : 479-89.
- Fishman, S.S., (1963). Studies on the excretion and distribution of radioactive nicotine and some evidence on the urinary metabolites, *Arch. Int. Pharmacodyn. Ther.*, 145 : 123-36.
- Greep, R.O. and Deane, H.W., (1949). Cytochemical evidence for the cessation of hormone production in the zona glomerulosa of the rat's adrenal cortex after prolonged treatment with desoxycorticosterone acetate, *Endocrinology*, 40 : 417-25.
- Haines, C.F.J.R., Mahajan, D.K. and Vesell, E.S., (1974). Radioimmunoassay of plasma nicotine in habitual and naive smokers, *Clin. Pharmacol. Ther.*, 16 : 1083-89.
- Hansson, E. and Schmitterloew, C.G., (1962). Physiological disposition and fate of ¹⁴C label nicotine in mice and rat, *J. Pharmacol. Exp. Ther.*, 137 : 91-102.
- Hill, A.B., (1977). *A Short Textbook of Medical Statistics*, 10th ed., London ELBS.
- Ingle, D.J., (1951). *Ann. Intern. Med.*, 35 : 652-72.
- James, C., (1980). Lewis's *Pharmacology*, 5th ed., edited by Churchill Livingstone. Edinburgh.
- Keele, C.A. Neil, E. and Joels, N., (1982). Adrenal cortex, in Samson Wright's *Applied Physiology*, 13th ed., Oxford, Oxford University Press.
- Kershbaum, A. and Shafida, H. (1968) Effects of smoking and nicotine on adrenocortical secretion, *JAMA*, 203 : 113-16.
- Kornovksy, M.L. and Dean, H.W., (1955). Aldehyde formation in the lipid droplets of adrenal cortex during fixation, as demonstrated chemically and histochemically, *J. Histochem. Cytochem.*, 3 : 85-102.
- Leeson, P.S. and Silvette, H., (1971). Tobacco: Experimental and clinical studies, *Suppl. (II)*, 172-173 Baltimore.
- Luna, H.T. and Lee, G., (1986). *Manual of Histologic Staining Methods of Armed Forces Institute of Pathology*, 3rd ed., New York McGraw-Hill Book Co.
- Livanon, T. Ferriman, D. and Corns, N., (1968). Hypothalmo-pituitary adrenal function in hyper corticosteroid states, *Acta. Endocrinol.*, 58 : 497-501.
- Moon, H.D., (1961). The adrenal cortex. New York, Paul B. Hoeber, Inc.
- Murphy, P.J. (1973) Enzymatic oxidation of nicotine to nicotine-1 (5)-immunon, *J. Biol. Chem.*, 248 : 2796-2800.

- Nishikawa, M., Morone, I. and Sato, T., (1963). Electron microscopic investigations of adrenal cortex, *Endocrinology*, 72 : 197-209.
- Putney, J.W. and Borzelleca, J.R., (1971). On the mechanism of ¹⁴C-nicotine distribution in rat maxillary gland *in vitro*, *J. Pharmacol. Exp. Ther.*, 178 : 180-91.
- Rubin, R.P. and Warner, W., (1975). Nicotine induced stimulation of steroidogenesis in adrenocortical cells of cat, *Br. J. Pharmacol.*, 53 : 357-62.
- Seifert, J. and Seifert, E. (1984) The influence of chronic nicotine administration on aldosterone and corticosterone plasma concentrations, *Klin Wochenshrm.*, 62 : 81-85.
- Silvette, H., (1961). Action of tobacco smoking on the adrenal medulla, *Arch. Intern., Med.*, 107 : 915-36.
- Stachenko, J. and Girond C.J.P., (1959). Functional zonation of adrenal cortex, site of ACTH action, *Endocrinology*, 64 : 7456-52.
- Suzuki, T., Ikeda, H. Narita, S. and Shibata, O., (1973). Adrenal cortical secretion in response to nicotine in conscious and anaesthetized dogs, *Quart. J. exp. Physiol.*, 58 : 139-42.
- Symington, G., (1962). *The Human Adrenal Cortex*, University of Glasgow, Glasgow.
- Turner, D.M., (1969). The metabolism of (¹⁴C) nicotine in the rat, *Biochem. J.*, 45 : 889-96.
- Vogt, M., (1951). Cortical secretion of isolated perfused adrenal, *J. Physiol.*, 113 : 124-56.
- Weizenecker, R. and Deal, W.B., (1970). Tobacco cropper sickness, *J. Fla. Med. Assoc.*, 57 : 13-14.
- Wennmalm, A., (1978). Effects of nicotine on cardiac prostaglandins and platelet thromboxane synthesis, *Br. J. Pharmacol.*, 64 : 554-63.
- Wood, E.F. and Richardson, J.A., (1961). *Tobacco Experimental and Clinical Studies*, Baltimore, Williams and Wilkins.